LECTURE 1 · WEDNESDAY 28 OCTOBER 2026

Meet AI

What AI actually is, the main tools you can use, and how to get access.

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Who am I?

ENGINEER

B.Sc. (Hons) Computer Engineering, University of Malta.

RESEARCHER

Research Support Officer on Project M.AI.ESTRO, Faculty of ICT: AI and virtual reality that guide people learning hands-on skills.

DAILY AI USER

I use AI every day for writing, coding and planning. These slides were built with AI help, and every slide was checked and decided by me.

You do not need to be an engineer for this course. That part is my job.

How this course works

BRING A LAPTOP

Every week. We work hands-on with real tools.

WI-FI: EDUROAM

Log in with your UM IT account. Setup guide: um.edu.mt/itservices/facilities/eduroam

ATTENDANCE

It is the only assessment. Eight Wednesdays, 12:00 to 14:00.

AFTER EACH LECTURE

Slides and revision notes go on the VLE. Short break at 13:00.

28 Oct	**Meet AI**: what AI is, the main tools and how to get access.
04 Nov	**Tokens and better results**: how AI reads your words, and how to ask well.
11 Nov	**Research and documents**: search, summarise, compare and check.
18 Nov	**Writing and communication**: emails, reports and messages for any audience.
25 Nov	**Planning, meetings and your own assistant**: plans, minutes, and building a Gem.
02 Dec	**Presentations and visuals**: structure, slides, speaking notes and images.
09 Dec	**Tables and data**: ask questions of a spreadsheet, no coding.
16 Dec	**Your AI workflow**: a repeatable process for your own field.

By December, you will be able to…

1
Pick the right type of AI tool for a task.

2
Write clear instructions that get useful answers.

3
Use AI for research, writing, planning, presentations and data.

4
Improve answers instead of accepting the first one.

5
Build reusable prompts and workflows for your field.

6
Know when to check, ask an expert, or decide yourself.

BEFORE WE START

Who's in the room?

Hands up, and I'll count. There are no wrong answers.

CLASS SURVEY

How often do you use AI tools?

Never	0
Once or twice	0
Every week	0
Every day	0

CLASS SURVEY

Which have you used? Hands up for each one.

ChatGPT	0
Gemini	0
Claude	0
Microsoft Copilot	0
Something else	0

CLASS SURVEY

Do you pay for an AI tool?

Free only	0
Paid plan	0
Through work or uni	0
Not sure	0

CLASS SURVEY

What have you used it for?

Study	0
Work	0
Everyday life	0
Not yet	0

CLASS SURVEY

Which area is your degree closest to?

Business	0
Health and care	0
Justice and social work	0
Design and arts	0
Science and education	0

CLASS SURVEY

How do you feel about AI?

Excited	0
Curious	0
Sceptical	0
Worried	0

PART A

What AI actually is

No maths and no coding. Just a clear picture of what is going on inside.

Three common myths

MYTH

“It knows everything.”

It learned patterns from text. There is no list of facts inside to look up.

MYTH

“It searches the internet for the answer.”

A chatbot writes from what it learned. Some tools can also search, but that is an extra feature.

MYTH

“It thinks like a person.”

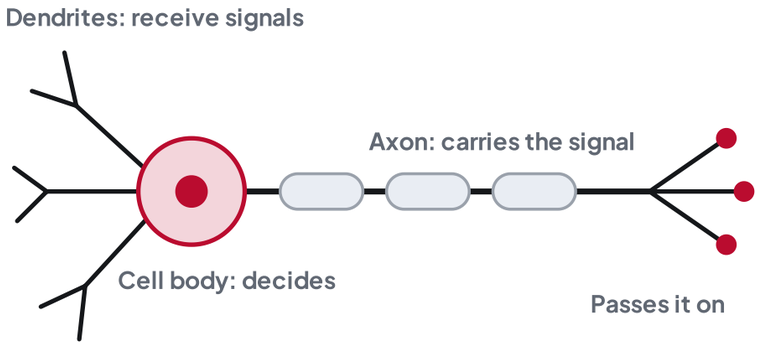
It predicts likely words. Very useful, but different from thinking.

IN ONE SENTENCE

AI chatbots learn patterns from huge amounts of text, then predict what comes next.

Hold on to this sentence. Everything today builds on it.

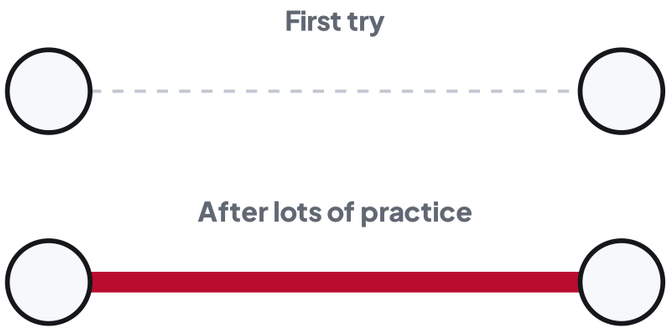
Start with your brain



- About **86 billion** neurons.
- Each one connects to **thousands** of others.
- A neuron collects signals. If they are strong enough, it **fires** and passes a signal on.

Neuron count: Azevedo et al. (2009), *Journal of Comparative Neurology*.

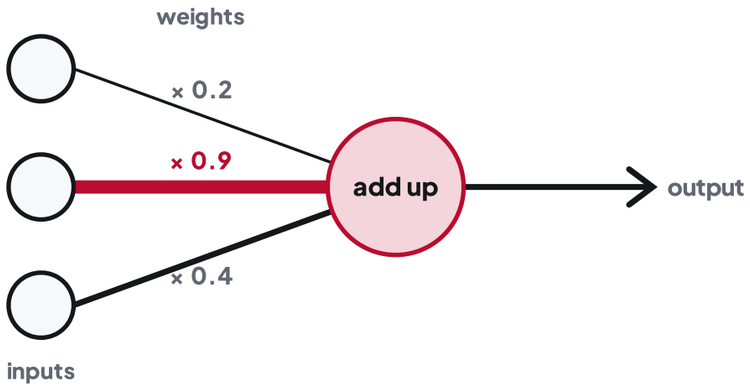
Learning = stronger connections



- Practise something, and the connections you use get **stronger**.
- That is why the hundredth bike ride is easier than the first.
- Neuroscientists sum it up as: **neurons that fire together, wire together**.

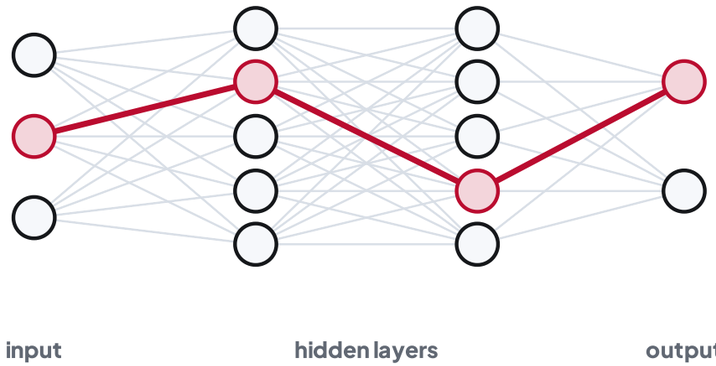
Idea from Donald Hebb, *The Organization of Behavior* (1949).

An artificial neuron: a simple maths copy



- It takes in numbers.
- Each input has a **weight**: how much it matters.
- It adds them up and passes a result on.
- The weights play the role of **connection strength**.

Connect lots of them: a neural network



- Layers of artificial neurons pass numbers forward.
- Big models have **hundreds of billions** of weights. Meta's open model Llama 3.1 has 405 billion.
- Your brain has roughly **100 trillion** connections: still far more.

Meta, *The Llama 3 Herd of Models* (2024). Synapse count is a common estimate; figures vary.

How it learns: guess, check, adjust

1
Show it some text with the next word hidden.

2
It guesses the missing word.

3
Compare the guess with the real word.

4
Nudge the weights so the right word becomes a little more likely.

Repeat this trillions of times. Nobody types in the rules: the patterns come from the examples.

A bit like a child learning a language

A CHILD

- Hears thousands of sentences.
- Nobody teaches grammar rules first.
- Starts to predict what people will say.

AN AI MODEL

- Reads trillions of words.
- Nobody programs the rules.
- Learns which words tend to follow which.

One big difference: a child also sees, touches and lives in the world. The model only has text.

GAME

Finish the phrase

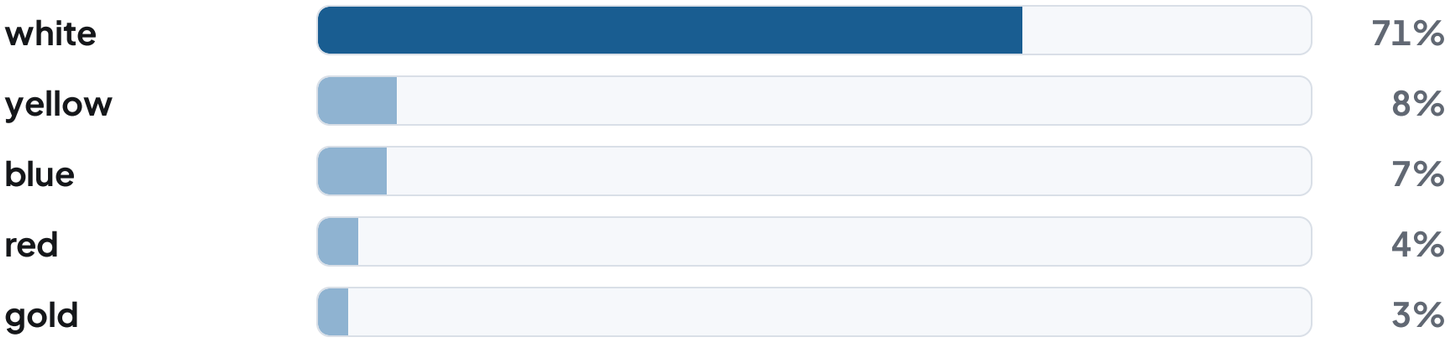
Shout out the first word that comes to mind.

FINISH THE PHRASE

Black and _____

WHAT A MODEL MIGHT PREDICT

Black and …



Illustrative numbers, to show the idea

Most of you said **white**. Your brain picked the most familiar continuation. That is a prediction.

FINISH THE PHRASE

Beauty and the _____

WHAT A MODEL MIGHT PREDICT

Beauty and the …



Illustrative numbers, to show the idea

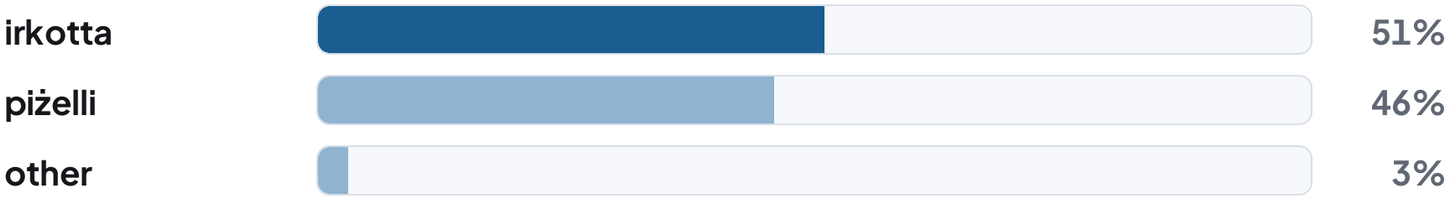
When one continuation dominates everything we have heard, the prediction is almost certain.

FINISH THE PHRASE

Pastizzi tal-_____

WHAT A MODEL MIGHT PREDICT

Pastizzi tal- …



Illustrative numbers, to show the idea

Two answers are almost tied. The model picks one, and if you ask again it may pick the other. That is why AI can give **different answers to the same question**.

FINISH THE PHRASE

The capital of Malta is

WHAT A MODEL MIGHT PREDICT

The capital of Malta is …



Illustrative numbers, to show the idea

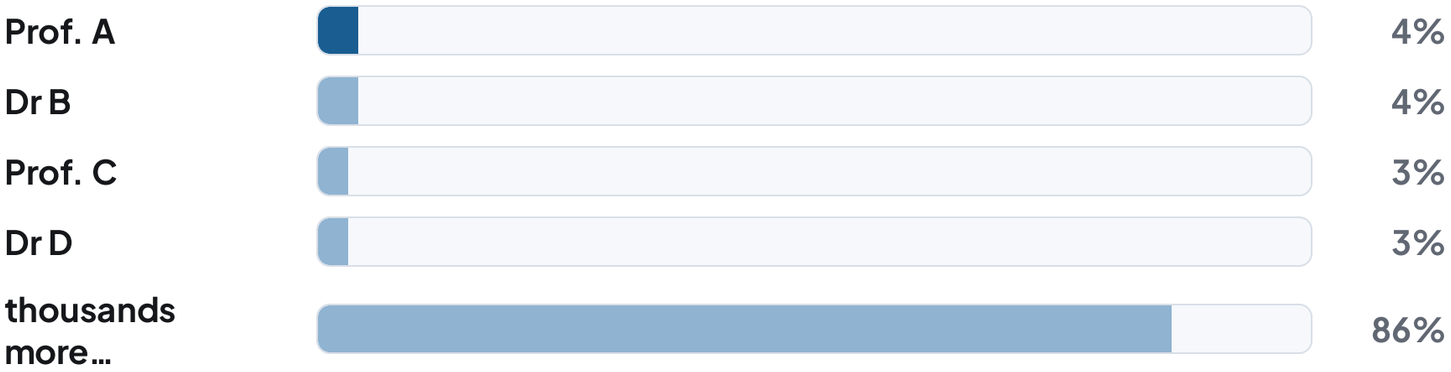
Facts that appear thousands of times come out right. But they are still **predictions**, not looked up.

FINISH THE PHRASE

The best lecturer at the University of Malta is

WHAT A MODEL MIGHT PREDICT

The best lecturer at the University of Malta is …



Illustrative numbers, to show the idea

There is no real answer, but the model will still write a fluent, confident one. This is where **made-up answers** come from.

What the game shows

YOUR BRAIN PREDICTS TOO

You finished every phrase without thinking about it.

AI DOES IT AT SCALE

One piece of a word at a time, again and again, to write whole answers.

LIKELY IS NOT TRUE

Usually right for common facts. Confidently wrong for rare ones.

SCALE

Meta's Llama 3 learned from about 15 trillion pieces of text.

Reading non-stop at 250 words a minute, 24 hours a day, a person would need roughly **85,000 years** to read that much.

Meta AI, Llama 3 announcement (April 2024): over 15 trillion tokens. Reading time is a rough estimate at about 0.75 words per token.

Why it sounds so sure of itself

WHAT IT IS TRAINED TO DO

Produce text that is **likely** and **sounds right**.

WHAT IT IS NOT BUILT TO DO

Know whether a statement is **true**.

So it is excellent at drafting, explaining and rewording. It still needs you to check facts, numbers and sources.

One more thing: it does not read words

AI splits text into **tokens**. Common words are one piece; rarer words are cut into chunks.

ENGLISH: 9 TOKENS

`Un` `bel` `ievable` ` results` ` at` ` the` ` University` ` of` ` Malta`

MALTESE: 2 WORDS, 6 TOKENS

`Past` `izzi` ` tal` `-` `irk` `otta`

Split with OpenAI's o200k tokenizer, used by recent ChatGPT models.

Next week: how tokens explain cost, usage limits, and why very long chats go wrong.

Part A in three sentences

1

AI chatbots are trained on huge amounts of text to predict what comes next.

2

What they learned is stored as billions of weights, not as a database of facts.

3

Likely is not the same as true. You stay the checker and the decision-maker.

BREAK

5 min

Back at 13:05. Next: the big three AI companies, their tools, and how to get access.